

WELL COMPLETION REPORT (SYNTHETIC)

OIL-MRN-011 - Moran-011

NWIS prototype document - not an Oil India Limited operational record

API Number	OIL-ASM-000123
Field / District	Moran / Dibrugarh
Block	PEL-ASSAM-03
Basin	Assam-Arakan
Formation	Tipam (TIP)
Spud / Completion	2014-01-08 / 2014-06-08
Measured / TVD	2155 m / 2120 m
Trajectory	Deviated; inc 26.0 deg; az 333.0 deg
Formation Interval	965-1543 m
Lithology	Fluvial sandstone with clay streaks
Pore Pressure Reference	1925 psi
Fracture Gradient	15.37 ppg
Mud System	PHPA Water-Based Mud
NWIS Well Risk	Moderate (54.5/100)

Casing Program

String	Set Depth (m)	Purpose
20 in conductor	80	Conductor
13 3/8 in casing	538	Surface
9 5/8 in casing	1249	Intermediate
7 in liner/casing	1896	Production

Significant Historical Events and Lessons

Date / Depth	Event	Severity	Evidence / mitigation / lesson
2014-03-11 998 m	Cementing Issue	Critical	Observed returns did not match calculated displacement. Mitigation: Use staged cementing response when returns deteriorate. Lesson: Volume imbalance is an important cement-placement warning.
2014-03-25 1180 m	Torque Spike	Critical	Torque rose without corresponding pressure anomaly. Mitigation: Reduce WOB/RPM and circulate before resuming normal parameters. Lesson: Avoid simultaneous high WOB and high RPM in correlated interval.
2014-05-28 2018 m	Kick	Critical	Drilling break followed by positive flow check. Mitigation: Increase mud weight within fracture-margin limits and monitor gas trend. Lesson: Escalate connection-gas trends before entering correlated pressure zone.
2014-02-15 669 m	Cementing Issue	High	Displacement pressure became unstable. Mitigation: Use staged cementing response when returns deteriorate. Lesson: Volume imbalance is an important cement-placement warning.
2014-03-17 1071 m	Kick	High	Flow-out exceeded flow-in and gas units increased. Mitigation: Shut in, verify influx and circulate using approved well-control procedure. Lesson: Treat pit gain plus gas increase as an early influx indicator.
2014-03-21 1130 m	Torque Spike	High	Sharp torque excursion occurred with RPM instability. Mitigation: Reduce WOB/RPM and circulate before resuming normal parameters. Lesson: Avoid simultaneous high WOB and high RPM in correlated interval.
2014-04-11 1403 m	Mud Loss	High	Intermittent circulation losses developed while drilling ahead. Mitigation: Spot graded LCM pill and circulate at reduced rate. Lesson: Prepare LCM package before approaching correlated offset-well loss depth.

Completion Summary

The synthetic well reached 2155 m MD in the Tipam interval. The NWIS knowledge record links 18 historical operational events to depth-matched telemetry. Risk classification is derived from formation, depth, trajectory and event history rather than formation alone.

This report is generated solely for SIH/NWIS prototyping and model development.